

Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE

**JC2 PRELIMINARY
EXAMINATION 2008**

H1 BIOLOGY

8875/1

Paper 1: Multiple Choice

12 September 2008

Friday

60 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer **all** questions. For each question, there are four possible answers. **A, B, C** and **D**.

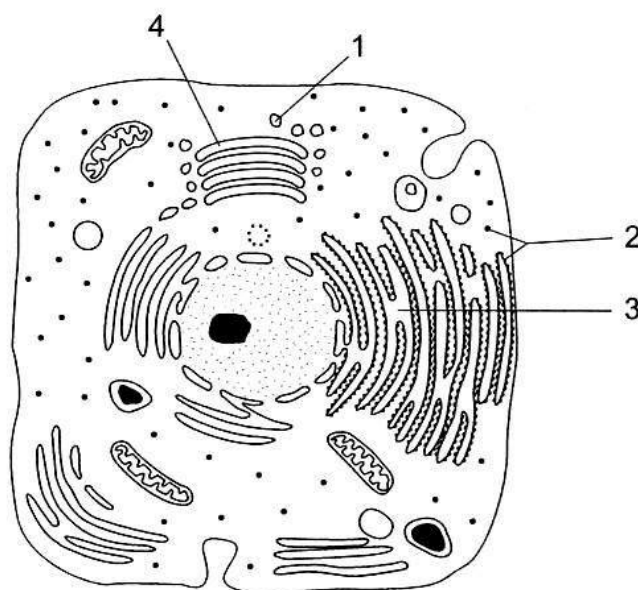
Choose the one you consider correct and record your choice in **soft pencil** on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in the booklet.

At the end of the examination, submit the multiple choice answer sheet only.

- 1 The diagram shows the ultrastructure of an animal cell.



What are the functions of the labeled structures?

	Synthesizing proteins from amino acids	Transporting proteins	Processing proteins	Secreting digestive enzymes
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

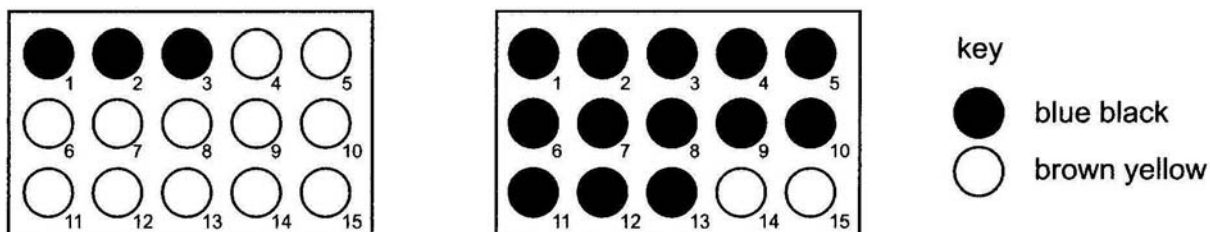
- 2 The primary structure of a molecule of enkephalin, a mammalian neurotransmitter is shown.

tyrosine – glycine – glycine – phenylalanine – leucine

Which combination correctly describes the synthesis and structure of one molecule of enkephalin?

	Number of molecules of water eliminated	Type of bond	Number of different R groups in molecule
A	4	Glycosidic	5
B	4	Peptide	4
C	5	Glycosidic	5
D	5	Peptide	4

- 3 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.

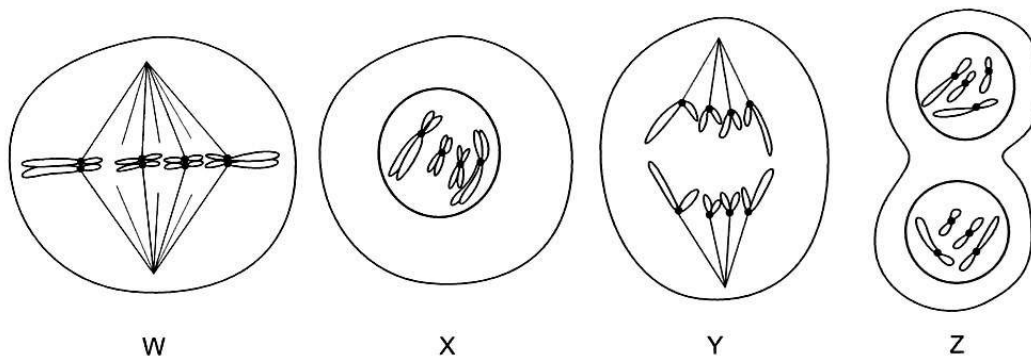


Which shows the correct temperature and times for the complete digestion of starch?

	Temperature / °C	Time / s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

- 4 Which environmental factor is associated with the development of melanomas (a type of skin cancer)?
- A Alcohol
 - B Tar
 - C Ultraviolet light
 - D X-radiation
- 5 Which of the following is **not true** about cancerous cells?
- A Ability to divide for a certain number of times.
 - B Ability to divide further when in contact with neighboring cells.
 - C Inability to differentiate properly.
 - D Inability to exhibit anchorage dependence.

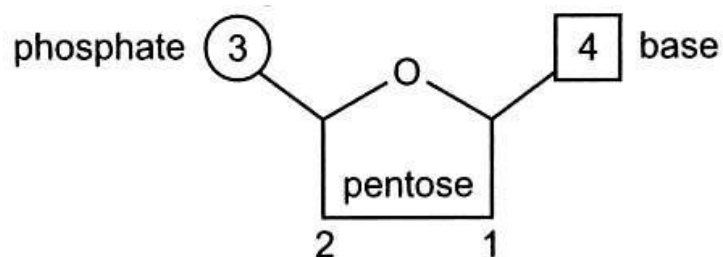
- 6 The diagrams show the cell cycle of a cell involved in asexual reproduction and the arrangement of two pairs of its chromosomes at different stages of the cycle.



Which combination represents the correct match between stage of cell cycle and arrangement of chromosomes?

	Stage of cell cycle	Arrangement of chromosomes
A	Prophase	Z
B	Anaphase	Y
C	Metaphase	X
D	Telophase	W

- 7 The diagram shows the structure of a nucleotide.



One type of polynucleotide consists of a single strand, twisted into a clover leaf shape. Which is the type of pentose of this molecule and the bonding between the nucleotides?

	Pentose	Bonding between nucleotides
A	Deoxyribose	Between positions 1 and 3
B	Deoxyribose	Between positions 2 and 3
C	Ribose	Between positions 4 and 4
D	Ribose	Between positions 2 and 3

- 8 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labeled. These bacteria (generation 1) were then grown in a medium containing only normal nitrogen (^{14}N) for two more generations (generations 2 and 3). The percentage of cells containing ^{15}N in each generation was estimated.

What will be the percentage of cells containing ^{15}N in generations 2 and 3?

	Percentage of cells containing ^{15}N	
	Generation 2	Generation 3
A	100	100
B	100	50
C	50	50
D	50	25

- 9 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- A** DNA ligase works only in the 3' to 5' direction.
 - B** The origins of replication occur only at the 5' end.
 - C** Helicases and single-stranded binding proteins work at the 5' end.
 - D** DNA polymerase can join new nucleotides only to the 3' end of a growing strand
- 10 In most organisms, all possible triplets of bases in DNA code for an amino acid, except for all the coding strand 'stop' triplets ATT, ATC and ACT. However in the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid, pyrrolysine.

The coding strand is complementary to mRNA. The **non-coding** (template) strand of a section of DNA in *M.barkeri* has the following sequence of bases.

TTT TTA TTG TAT TAC TAG TGT TAA TGA

How many amino acids can be made into a polypeptide by *M.barkeri* from the **coding** strand of this region of DNA?

- A** 5
- B** 6
- C** 7
- D** 9

- 11** The table shows the mRNA codons for 11 different amino acids.

Amino acid	mRNA codon	Amino acid	mRNA codon	Amino acid	mRNA codon
Ala	GCG	Lys	AAG	Arg	CGC
Glu	GAG	Pro	CCU	Phe	UUC
His	CAC	Thr	ACU	Gly	GGA
Leu	CUG CUC	Val	GUG		

The first seven DNA triplets coding for a protein are shown below.
These are DNA triplets from the coding strand, complementary to the mRNA.

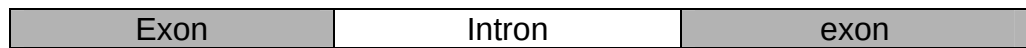
CAC GAG GCG AAG GGA CAC CGC

A mutation occurs when the seventeenth nucleotide in the DNA sequence is changed to G.

What is the amino acid sequence for the mutated section of DNA?

- A** His Glu Ala Lys Gly Arg Arg
 - B** Val Leu Arg Phe Pro Ala Ala
 - C** His Glu Ala Lys Gly His Arg
 - D** Val Leu Arg Phe Pro Val Ala
- 12** Which feature distinguishes eukaryotic genomes from prokaryotic genomes?
- A** Eukaryotic genomes contain non-coding sequences called introns.
 - B** Eukaryotic genomes contain no short repetitive sequences of DNA.
 - C** Eukaryotic genomes have genes aggregated into operons.
 - D** Eukaryotic genomes possess plasmids.
- 13** A regulator protein binds to a site on an mRNA molecule and covers the triplet AUG. What is the result of this?
- A** Transcription of the mRNA is prevented.
 - B** Transcription of the mRNA is promoted.
 - C** Translation of the mRNA is prevented.
 - D** Translation of the mRNA is promoted.

- 14** The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect.

- A** Activation of the gene by a steroid hormone.
 - B** Attachment of mRNA to a ribosome.
 - C** Binding of RNA polymerase to DNA.
 - D** Splicing of pre-mRNA.
- 15** The diagram shows the banding pattern of two human chromosomes. P is a normal chromosome.



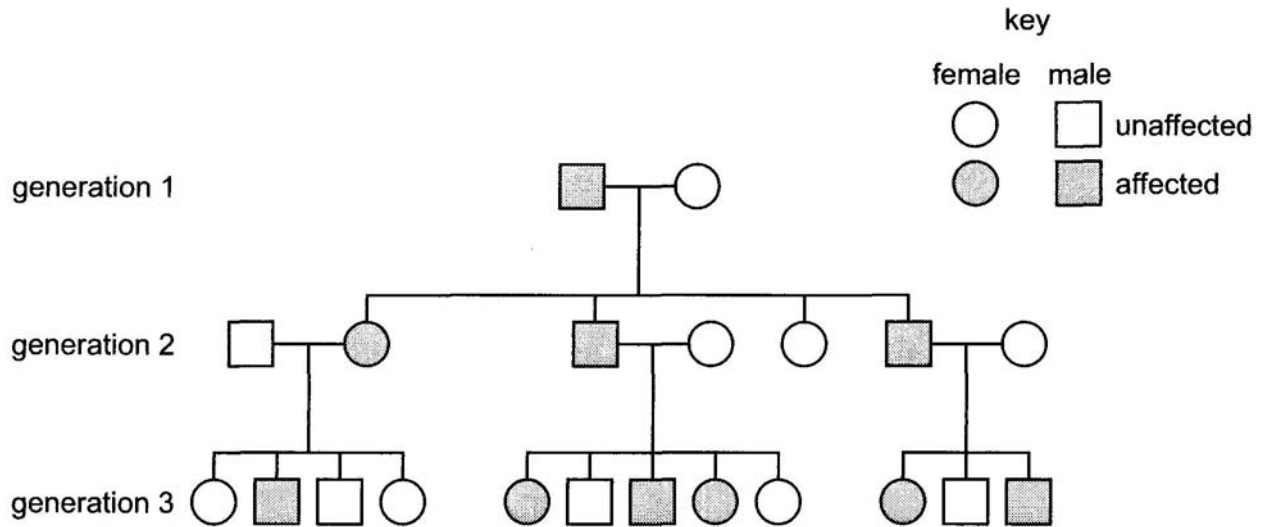
What accounts for chromosome Q?

- A** Crossing over between sister chromatids.
 - B** Inversion of part of the chromosome.
 - C** Deletion of part of the chromosome.
 - D** Translocation of part of another chromosome.
- 16** Some organisms change the composition of their membranes to maintain membrane fluidity when temperature changes.

Which changes occur in the composition of the membranes following a change to low temperature?

- A** Increase phospholipids, decrease cholesterol.
- B** Increase saturated fatty acids, increase cholesterol.
- C** Increase saturated fatty acids, decrease cholesterol.
- D** Increase unsaturated fatty acids, increase cholesterol.

- 17 The pedigree shows the inheritance of a genetic condition in a family for three generations.



Which evidence indicates that this genetic condition is autosomal?

- A Affected females always have affected sons.
 - B Affected males do not pass it on to their sons.
 - C Affected parents always have affected offspring.
 - D Males and females are equally affected.
- 18 A cross between a round-leaved, tall plant and round-leaved, dwarf plant produced the following offspring:

121 round-leaved, tall plants
 121 round-leaved, dwarf plants
 42 oval-leaved, tall plants
 37 oval-leaved, dwarf plants

Key
 R – round leaf
 r – oval leaf
 T – tall
 t – dwarf

What were the genotypes of the parents?

- A RrTt X Rrtt
- B RrTt X RRtt
- C RrTT X Rrtt
- D RrTT X RRtt

19 Which statements are true?

1. Continuous variation is always influenced by the environment.
2. Continuous variation is always caused by many genes.
3. Discontinuous variation is always caused by one gene.
4. Discontinuous variation is sometimes influenced by the environment.

- A** 1 and 4
B 2 and 3
C 2 and 4
D 3 and 4

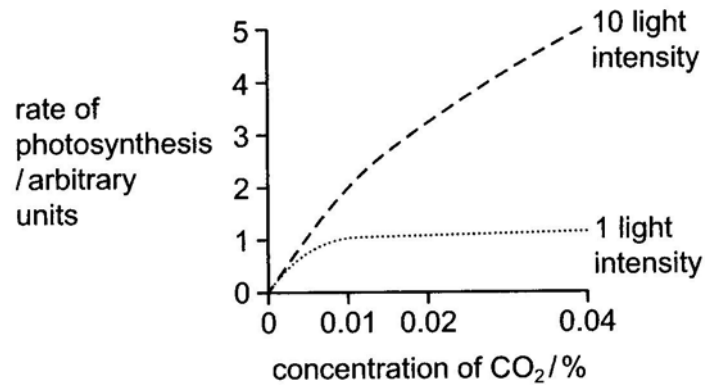
20 In addition to ethanol, what does the fermentation of glucose by yeast yield?

	Number of molecules		
	CO ₂	Reduced NAD	ATP
A	0	0	36
B	0	4	2
C	2	0	2
D	2	4	36

21 Which combination correctly describes light-dependent reactions of photosynthesis?

	Site of reaction in chloroplast	ATP	NADP
A	Stroma	ATP --> ADP + Pi	Oxidized
B	Granum	ATP --> ADP + Pi	Reduced
C	Stroma	ADP + Pi --> ATP	Oxidized
D	Granum	ADP + Pi --> ATP	Reduced

- 22 The graph shows the effect of carbon dioxide concentration on the rate of photosynthesis, at two different light intensities.

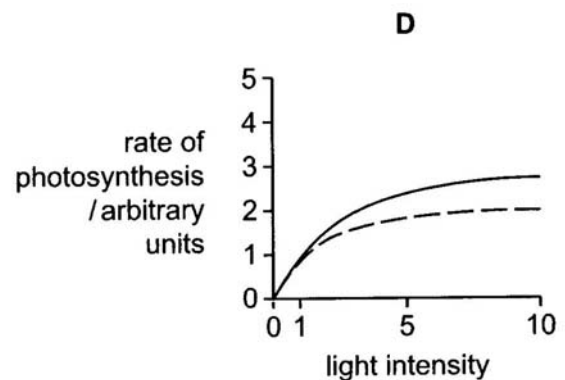
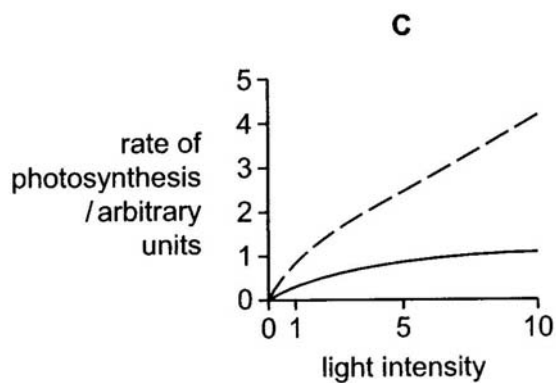
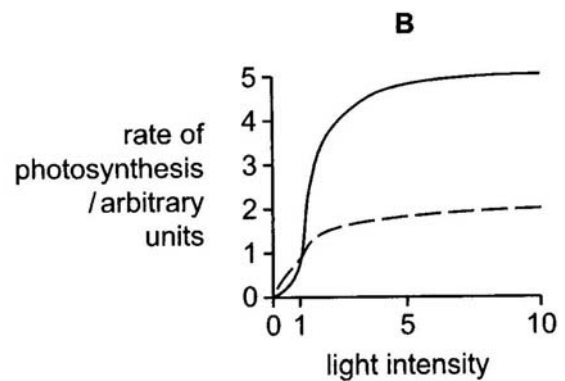
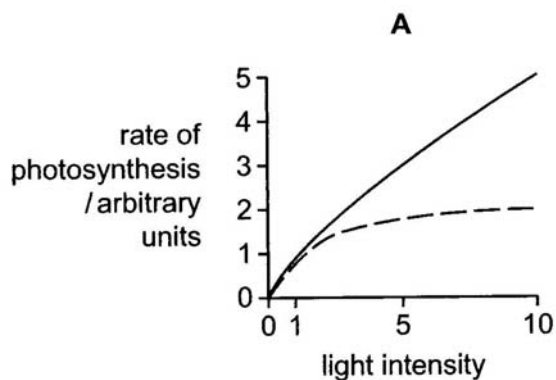


Which graph correctly shows the effect of light intensity on the rate of photosynthesis, at two different carbon dioxide concentrations?

key

— — — 0.01 % concentration of CO₂

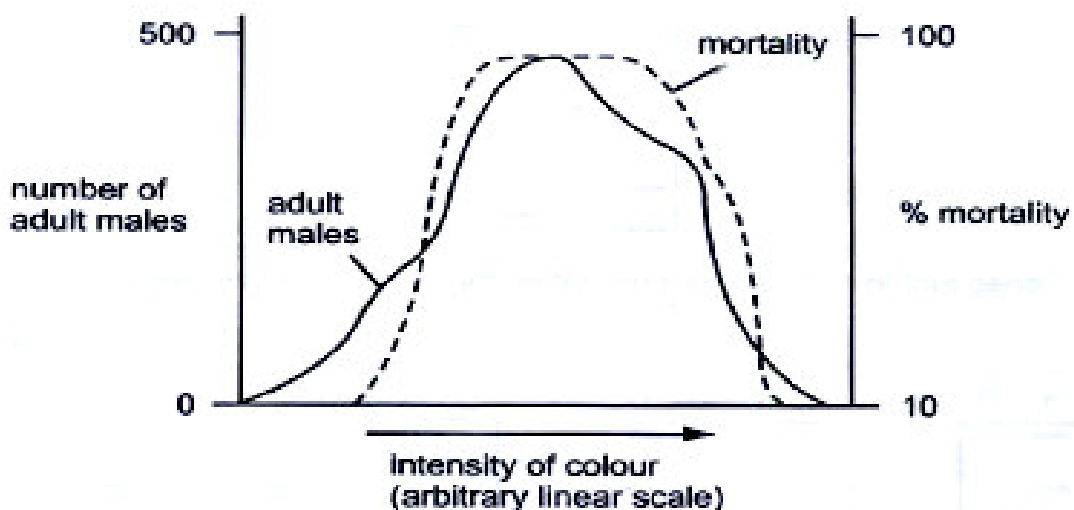
———— 0.04 % concentration of CO₂



- 23** In non-cyclic photophosphorylation, when the reaction centre of Photosystem II absorbs light, its electron is excited and accepted by an electron acceptor X before it is passed on to a series of electron carriers.

In the presence of an inhibitor to electron X, which of the following would take place?

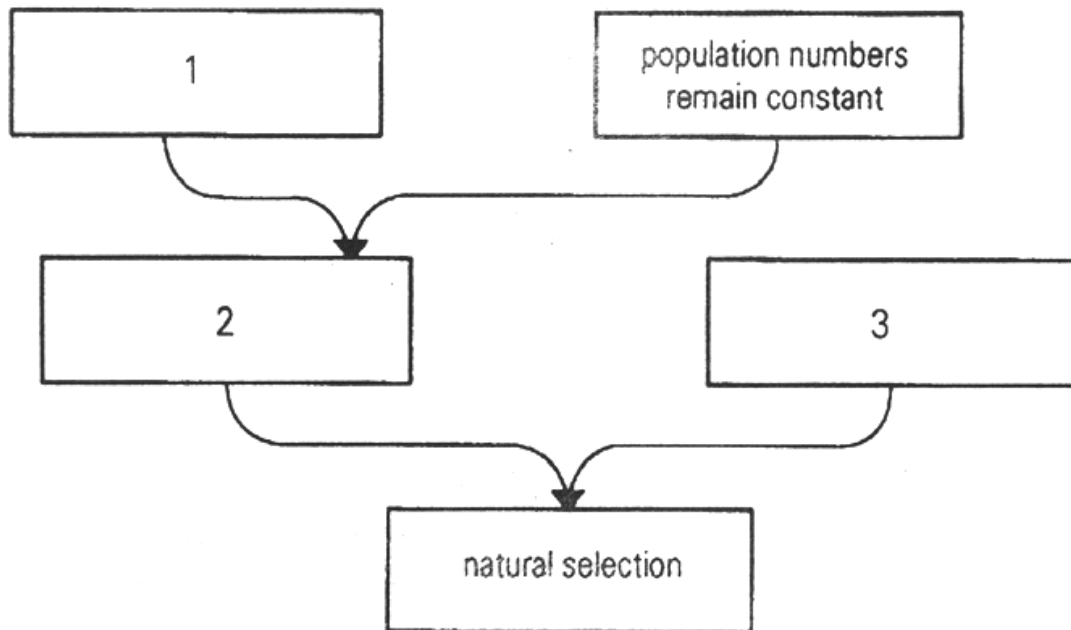
- A** No oxygen gas is produced.
 - B** No carbon dioxide gas is produced.
 - C** More ATP and reduced NADP are produced.
 - D** More glucose is produced.
- 24** The graph below shows data on a population of a species of moth which shows considerable variation in color intensity.



What conclusion can be made from this graph?

- A** Color variation is environmentally induced.
- B** Color variation is genetically determined.
- C** Extreme forms are favored by natural selection.
- D** The species shows discontinuous variation with respect to color.

- 25 The diagram shows a summary of Darwin's theory of natural selection.



Which statements should be placed in boxes 1, 2 and 3?

	There is a struggle for existence.	Variation is shown in all population.	Individuals show great reproductive capacity.
A	1	2	3
B	2	1	3
C	2	3	1
D	3	1	2

- 26 At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60°C.

What explains why the denatured DNA strands anneal with primers and **not** each other?

- A** The primers are shorted and anneal more easily.
- B** The primers anneal only to the 3' end of the denatured DNA.
- C** The primer concentration exceeds the denatured DNA concentration.
- D** The temperature prevents the denatured DNA from annealing together.

27 Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?

- A Control their own replication
- B High copy number
- C Small in size
- D Several restriction sites made by a restriction enzyme

28 The diagram shows a section of a DNA molecule.

ATCGTAGAATTCGGCTCGAGGCATTAGAATTCCTAGGATCGAATTGGCAT
TAGCATCTTAAGCCGAGCTCCGTAATCTTAAGGATCCTAGCTTAACCGTA

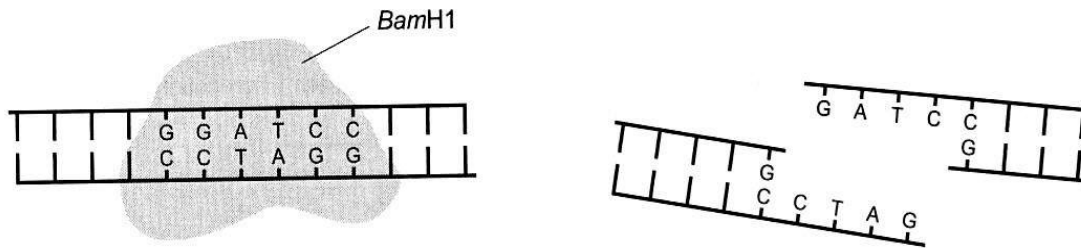
This molecule was cut under ideal conditions, using two different endonucleases, *EcoR1* and *Tac1*, which recognize different restriction sites:

EcoR1 GAATTC *Tac1* TCGA
 CTTAAG AGCT

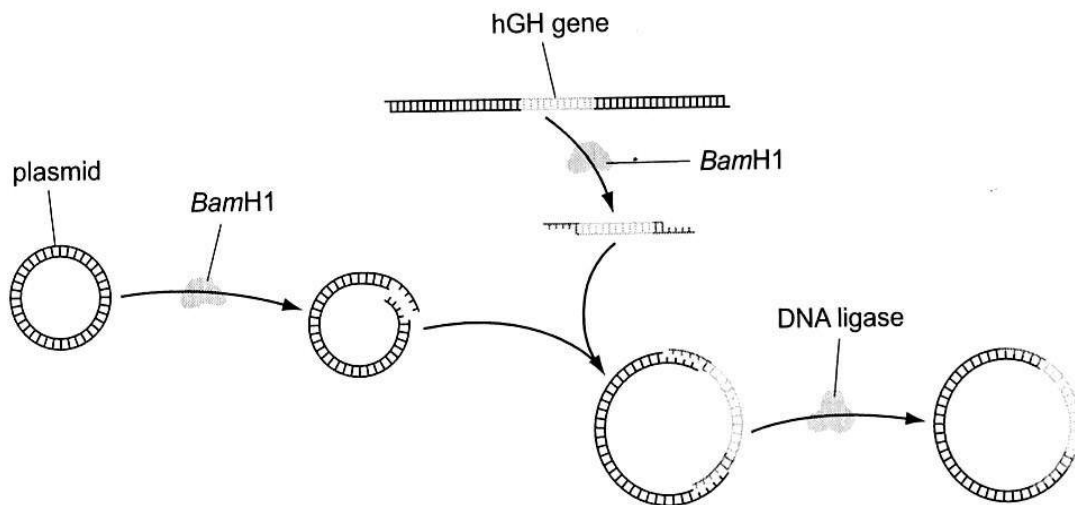
Some of the DNA fragments produced were spliced into a plasmid cut by *Tac1*. Which combination is correct?

	Number of DNA fragments	Number of fragments spliced
A	4	2
B	4	3
C	5	2
D	5	3

29 BamH1 is a restriction enzyme that cuts DNA as shown in the diagram.



The diagram below shows part of the procedure for producing E.coli that will synthesis human growth hormone, hGH.



At the end of this process, many plasmids do not contain the hGH gene. What could explain this?

- A** Different alleles of the hGH gene have different sticky ends.
- B** Not all the plasmids cut by BamH1 have sticky ends.
- C** Some of the plasmids are cut at more than one position.
- D** The sticky ends of some of the plasmids rejoin with each other.

30 Which statement describes a unique feature of stem cells in human?

- A** They can differentiate into different cell types.
- B** They can undergo meiosis.
- C** They have a complete human genome.
- D** They have multiple copies of active genes.